

Real-Time Urban Traffic Congestion Prediction Using Distributed Big Data Processing and Deep Learning

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Abstract— Urban traffic congestion has emerged as a major challenge impacting mobility, fuel efficiency, and environmental sustainability in rapidly expanding metropolitan areas. The proliferation of big data and real-time data streams generated by sensors, GPS devices, and Internet of Things (IoT) systems has created new opportunities for the development of intelligent traffic management solutions. This paper investigates the application of deep learning (DL) techniques for real-time traffic congestion prediction, emphasizing architectures such as Long Short-Term Memory (LSTM) networks, Gated Recurrent Units (GRU), and Transformer-based models. In contrast to traditional machine learning approaches, deep learning facilitates automatic feature extraction from complex spatiotemporal traffic data, thereby enhancing prediction accuracy, robustness, and scalability. Furthermore, the study highlights the significance of big data frameworks such as Apache Kafka and Apache Spark, which enable efficient real-time ingestion, processing, and analysis of high-velocity traffic data streams. This paper presents a comprehensive review of recent advancements in the field, encompassing benchmark datasets, deployment frameworks, existing challenges, and future research directions. The findings aim to support the implementation of deep learning-driven, real-time traffic prediction systems that form the foundation of next-generation smart city infrastructures.

Keywords: *Traffic Congestion Prediction, Deep Learning, Spatiotemporal Modeling, Long Short-Term Memory (LSTM), Transformers, Big Data Analytics, Real-Time Data Streams, Apache Spark, Apache Kafka*

I. INTRODUCTION

Rapid urbanization and population growth have significantly increased vehicular density, leading to frequent traffic congestion in metropolitan regions. Such congestion not only prolongs travel times but also escalates fuel consumption, air pollution, and economic losses. Conventional traffic control systems, which depend on predefined rules and classical statistical approaches, often struggle to adapt to the highly dynamic and non-linear nature of real-world traffic environments.

Recent advancements in big data analytics and deep learning (DL) have transformed the development of intelligent transportation systems (ITS). Massive streams of real-time traffic data are continuously produced from diverse sources such as inductive loop detectors, GPS-enabled vehicles, mobile devices, surveillance cameras, and IoT sensors. Although these heterogeneous data sources offer immense potential for predictive modeling, their high velocity, variety,

and volume present substantial challenges for data storage, processing, and real-time analytics.

To address these challenges, big data frameworks like Apache Kafka—for real-time data ingestion—and Apache Spark—for distributed stream processing—are increasingly adopted to build low-latency, high-throughput data pipelines. On top of these infrastructures, deep learning architectures such as Long Short-Term Memory (LSTM) networks and Transformer models have demonstrated exceptional capability in capturing complex, non-linear spatiotemporal dependencies in traffic flows, enabling accurate congestion prediction.

Compared to traditional machine learning approaches, deep learning offers enhanced feature extraction, scalability, and adaptability, especially when dealing with high-dimensional and unstructured data like sensor readings, traffic images, and GPS traces. For instance, LSTM networks effectively model sequential traffic data to provide accurate congestion predictions at the road-segment level.

This survey provides a comprehensive review of the state-of-the-art research in real-time traffic congestion prediction using deep learning and big data technologies. It discusses existing DL architectures, real-time data infrastructures, benchmark datasets, evaluation metrics, deployment strategies, and identifies key research gaps in the field. The objective is to emphasize how DL-driven frameworks can foster robust, scalable, and adaptive traffic management systems for emerging smart cities.

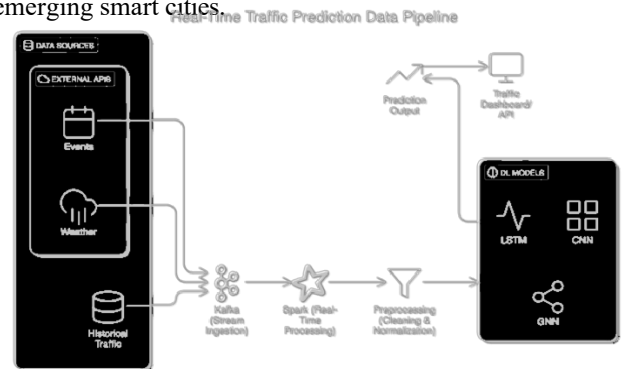


FIGURE 1. General Architecture for Real-Time Traffic Congestion Prediction Using Bigdata and Deep Learning

Big data platforms like Apache Kafka and Apache Spark have emerged as powerful tools for handling high-speed, real-time

traffic data streams. Coupled with advanced machine learning (ML) and deep learning (DL) algorithms—such as Support Vector Machines (SVM), Random Forest (RF), K-Nearest Neighbors (KNN), Linear Regression, and Long Short-Term Memory (LSTM) networks—these systems can uncover complex spatio-temporal patterns in traffic flow and forecast future congestion with high accuracy.

This survey aims to provide a comprehensive overview of the current state-of-the-art approaches in real-time traffic congestion prediction using big data and DL. We examine the architectures, algorithms, technologies, and applications that underpin modern intelligent traffic systems. Furthermore, we identify key challenges, research gaps, and potential directions for future work in building scalable, real-time, and AI-enhanced traffic prediction systems suitable for smart city infrastructures.

II. LITERATURE SURVEY

Noureen Zafar et al. [1] proposes a hybrid GRU-LSTM based deep learning model and applied it on city-wide novel traffic data integrated from heterogeneous sources. It devises indigenous data pipeline that is composed of a set of algorithms dealing with map matching, sparsity handling, outlier removal, zero speed adjustments, Open Street Map (OSM) and segment mapping etc. Extensive experimentations have been carried out to demonstrate the improved performance of the proposed method. It describes a mechanism to integrate multiple sources of data into a hybrid feature space. Basically, it utilizes an ETA based congestion index as a road network state evaluation indicator that distributes the traffic state primarily into three categories ranging from smooth to congested to Highly Congested class. It integrates the traffic load, GPS, weather, special conditions with the OSM data set and employed different deep learning and machine learning algorithms. Among classical learning techniques, Random Forest provided the best results whereas in deep learning algorithms, GRU-LSTM proved to be the best with the highest accuracy. Finally, it concluded that deep learning techniques are the most reliable learning techniques providing maximum accuracy and better yield than classical learning techniques when are applied in traffic congestion problems on arterial roads. These further paves way towards automatic labeling of the classes instead of using congestion indexes and automatic optimization of hyper parameters using adaptive techniques in future studies.

Jesus Mena-Orej et al. [2] conducted a comprehensive study comparing modern deep neural network architectures for traffic prediction. This work evaluates various advanced models under identical conditions using real-world traffic datasets, including scenarios of both normal flow and traffic congestion. The findings reveal that a model's performance in regular conditions does not necessarily reflect its effectiveness during emerging congestion scenarios. The study emphasizes the importance of assessing prediction models under congestion-specific circumstances. It also highlights the advantage of incorporating both spatial and temporal traffic dynamics over solely temporal analysis, especially for enhancing accuracy during congestion. Additionally, models that incorporate error feedback mechanisms, referred to as

error recurrent models, consistently outperformed those that do not, regardless of the traffic condition.

Saurabh Pratap Singh Rathore et al. [3] proposes an AI-driven system designed to predict and manage traffic congestion. The system leverages continuous traffic data from IOT devices, such as images, GPS, and inductive loop sensors, to monitor real-time traffic conditions. Traffic predictions are made using a combination of CNN and LSTM networks. Additionally, Geographic Information System (GIS) technology is integrated for spatial analysis and dynamic traffic control, enabling real-time traffic light adjustments and providing drivers with alternative routes to avoid congestion. Preliminary experiments show that intelligent traffic management systems can significantly improve traffic flow and reduce congestion in large cities. The paper also explores challenges related to data privacy, scalability, and future advancements, including the integration of self-driving vehicles and the application of reinforcement learning. hence, the traffic congestion prediction and management system using Artificial intelligence provided a solution to the increased congestion in traffic systems in urban areas. By integrating IoT sensor data feeds that display real-world scenarios, the proposed system can employ deeper and more advanced constructs such as CNN LSTMs for further and more precise traffic pattern prediction and, consequently, a more dynamic traffic flow control. The integration of GIS is also applied to improve spatial analysis and the ability to make improved routing decisions. At the same time, real-time data processing enables the system to adapt to signal timings and rerouting changes quickly.

Malathi et al. [4] presents an innovative approach to managing traffic congestion through an intelligent system that combines advanced congestion detection with adaptive signal control. The system leverages artificial intelligence to process various data sources, including real-time traffic flow, road conditions, and historical data, enabling it to detect congestion patterns with high accuracy. Machine learning algorithms are employed to analyze these patterns, allowing the system to automatically assess congestion levels and optimize signal timing dynamically based on factors such as traffic volume, queue length, and delay. The effectiveness of this approach has been validated through simulation studies, where the system demonstrated impressive performance metrics: an accuracy of 94.89%, a precision of 97.40%, a recall of 96.38%, and an F1 score of 96.76%. These results highlight the system's capability to effectively manage traffic conditions by accurately detecting congestion and minimizing false positives. By integrating these advanced techniques, the proposed solution significantly enhances traffic flow, reduces travel time, and improves the overall driving experience. The Intelligent Traffic Congestion Identification and Signal Switching System presents a groundbreaking solution to the growing challenge of traffic congestion. By integrating advanced congestion detection techniques with adaptive signal control, this system offers a novel approach to mitigating traffic issues. The utilization of artificial intelligence enables real-time analysis of traffic patterns, allowing for the accurate identification of congested areas, which is essential for efficiently adjusting signal timing to optimize traffic flow. The system's adaptability and automation make it an effective and

practical tool for managing congestion, thereby improving overall transportation efficiency and reducing travel times for commuters.

Hongsuk Yi et al. [5] developed an automated machine learning framework aimed at optimizing hyperparameters for traffic prediction models, particularly focusing on major highway systems. Recognizing the challenges associated with training deep learning models-such as the time and expertise required for hyperparameter tuning-they introduced the HyperNet framework. This framework leverages advanced techniques like Bayesian optimization and meta-learning to streamline the hyperparameter search process. By integrating HyperNet with Long Short-Term Memory (LSTM) networks, the study effectively captures temporal variations in traffic data, enhancing the prediction of highway congestion. The primary objective is to automate the training process, reducing reliance on manual configuration and improving model performance across diverse traffic datasets.

Humayun Kabir et al. [6] proposed an Intelligent Traffic Management System (ITMS) utilizing the Internet of Things (IoT) to tackle traffic congestion, a significant hurdle for modern cities. The ITMS gathers real-time traffic data (vehicle count, speed, lane occupancy) through strategically positioned sensors at intersections. A central control system analyzes this data and dynamically adjusts traffic signals, allocating green time based on actual traffic flow. This approach aims to optimize traffic flow, significantly reduce congestion, and improve efficiency. The paper explores the potential benefits of the proposed system, including shorter travel times, improved air quality, and enhanced safety for pedestrians and vehicles. The study proposes an intelligent technique for regulating and improving traffic signal systems that react in real time to automobile traffic. Preventing congestion and ensuring traffic flow requires controlling intersections in densely populated regions. For a variety of signal cycles, it evaluates both the fixed and improved traffic lights.

Mouna Zouari Mehd et al. [7] proposed a differential-entropy-based approach, it is a supervised congestion prediction from traffic meta-parameters based on a convolutional neural network was proposed. Traffic parameters includes node localization, date, day of the week, time of day, special road conditions, and holidays. The proposed model is validated on the CityPulse dataset, which is a set of vehicle traffic records. Simulation results on the CityPulse dataset illustrate that the proposed approach yields accurate prediction rates for different nodes considered. The proposed system can prevent traffic congestion by reorienting the drivers to follow other itineraries. The simulation results illustrate that the proposed approach yields accurate prediction rates for the different nodes considered.

Kavitha et al. [8] introduced an IoT-based traffic control system designed to alleviate this issue by using RFID technology and Arduino microcontrollers. The system detects emergency vehicles via RFID tags and dynamically adjusts traffic signals to provide an uninterrupted path. Implemented with RFID receivers and Arduino controllers, the system ensures real-time management of traffic signals. Results from preliminary tests indicate a substantial reduction in transit

times for emergency vehicles. This study demonstrates the potential of IoT-based solutions in improving urban traffic management, enhancing both public safety and emergency response efficiency. The integration of RFID technology and Arduino controllers enables real-time dynamic adjustments to traffic signals. While the system demonstrates considerable improvements, it has certain limitations. Potential interference issues from other RFID equipped vehicles and the dependency on the reliability of the RFID and Arduino components need to be addressed.

Yuan-yuan Chen et al. [9] discussed online open data to provide traffic related information. Traffic conditions collected from web-based map services are used to demonstrate the feasibility. The stacked long short-term memory model, a kind of deep architecture, is used to learn and predict the patterns of traffic conditions. Experimental results show that the proposed model for traffic condition prediction has superior performance over multilayer perceptron model, decision tree model and support vector machine model. Online open data are potential sources to be used to provide traffic related information. This paper reviews four categories of open data platforms, i.e., web map, social media, local events and weather website. Traffic condition data collected from web map provider to learn the hidden patterns of traffic conditions. It proposes a deep learning approach with a stacked LSTM model for traffic condition prediction. Online training was applied during the test phrase. It was compared with performance of the proposed stacked LSTM model with MLP model, decision tree model and SVM model. Experiments show the proposed method is superior to the competing methods.

Leixiao Li et al. [10] proposed the MF-TCPV, it uses Data-X and DataV to implement the integration of multi-source heterogeneous traffic data and the visualization of congestion prediction results. A deep prediction model named LSTM-SPRVM based on deep learning algorithms, machine learning algorithms, and Spark parallelization technology for the prediction of traffic congestion features in the future is proposed. In MF-TCPV, traffic congestion is divided into six levels based on Fuzzy Comprehensive Evaluation and traffic congestion features such as average speed, road occupancy rate, and traffic flow density. The experimental results demonstrate that MF-TCPV is capable of predicting the traffic congestion accurately and displaying prediction results visually. LSTM-SPRVM is better than other existing deep learning models in terms of prediction accuracy, and MF-TCPV can intuitively visualize the prediction results of traffic congestion.

Naveen et al. [11] proposed a model that has for predicting and visually representing congestion of traffic using machine learning as well as deep learning. This initiative with the purpose of examining traffic data to obtain insights for traffic regulation and planning. The process initiates with an initial exploration of data through an Exploratory Data Analysis (EDA) and to perceive the distribution and patterns within the dataset. Descriptive statistics, correlation analysis, and visual aids are employed for this purpose. Following the EDA phase, tactics for manifold learning are utilized to simplify the dataset while retaining crucial information. Identifying

significant features influencing the prediction of traffic volume is facilitated by methods for feature importance and

selection. The obtained features are rigorously evaluated to comprehend traffic patterns and traffic jam levels. Visual representations capture the distribution of traffic volume based on factors like time, day, and month, assisting in pinpointing congestion hotspots and optimizing traffic flow. This project underscores the powerful impact of data analysis and feature extraction in offering practical insights for traffic regulation strategies. Route Rover, with its machine learning driven approach to transportation routing, offers potential benefits such as enhanced travel efficiency and reduced congestion. Yet, its success hinges on the fidelity of real-time data and scalability considerations. With ongoing refinement, Route Rover could become a transformative tool for optimizing travel routes and mitigating traffic challenges.

Supriya Tel sang et al. [12] explored a novel method to traffic control that enhances signal control systems in metropolitan environments by applying machine learning techniques. This research revolves around a methodology that improves vehicle flow and reduces traffic congestion. This paper elaborates on the methodology utilized for collection of data and vehicle detection methods used for optimizing traffic signals. It incorporates machine learning algorithms into traffic signal control systems offers a promising solution for enhancing urban traffic management. This model adapts signal timings dynamically, resulting in substantial improvements in overall traffic efficiency. Additionally, the system's ability to prioritize emergency vehicles highlights its potential to improve public safety and response times during critical situations. The successful implementation of our algorithm emphasizes the significance of innovative traffic management and the revolutionary effects of machine learning technology on the development of future metropolitan planning and infrastructure development.

III. METHODOLOGY

The objective of the system is to design a robust framework for real-time traffic congestion prediction using various deep learning architectures. Unlike conventional methods that rely on static datasets, this approach integrates API-based live traffic data acquisition and dynamic model evaluation. The methodology consists of five key stages: Data Acquisition, Preprocessing, Feature Engineering, Model Design & Training, and Evaluation

A. Data Acquisition:

Traffic-related data is collected in real-time from publicly available APIs such as TomTom. These API provide dynamic information including:

- Current speed S_c
- Free flow speed S_f
- Travel time T
- Geographic coordinates

In cases where a congestion index C_i is not provided, it is computed using the formula

$$C_i = 1 - \frac{S_c}{S_f}$$

Where:

- S_c = current observed speed
- S_f = free-flow or Ideal speed
- $C_i \in [0, 1]$, with higher values indicating greater congestion

B. Data Preprocessing

The acquired data is structured and cleaned to remove noise and inconsistencies. The following steps are applied:

1. Missing Value Handling: Forward fill or linear interpolation is used.
2. Normalization: Features are scaled using Min-Max normalization:

$$x_{norm} = \frac{x - x_{min}}{x_{max} - x_{min}}$$

3. Timestamp Alignment: Timestamps are converted to a uniform time zone and format.
4. Sequence Framing: A sliding time window approach is used to create sequences for temporal prediction. For a window of size w :

$$X_t = \{x_{t-w+1}, \dots, x_t\}, \quad y_t = C_{i,t+1}$$

C. Feature Engineering

Along with primary features, derivative features such as speed variability, hour of the day, weekday/weekend flag are considered to enrich model input.

$$Speed\ Variability = \frac{\sigma(S_c)}{\mu(S_c)}$$

Where:

- $\sigma(S_c)$ is the standard deviation of current speed
- $\mu(S_c)$ is the mean speed over the window

D. Deep Learning Models Explored

Multiple deep learning models are evaluated to determine the most effective architecture for predicting congestion in real time. These include:

1. Long Short-Term Memory (LSTM)
2. Gated Recurrent Unit (GRU)
3. Random Forest
4. Hybrid Models (e.g. LSTM-GRU, GRU-LSTM)

Each model is trained using the processed time-series data to predict either:

- A continuous congestion index
- Or a congestion class (e.g., low, medium, high)

The training loss function depends on the output type:

- For Regression

$$L_{MSE} = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2$$

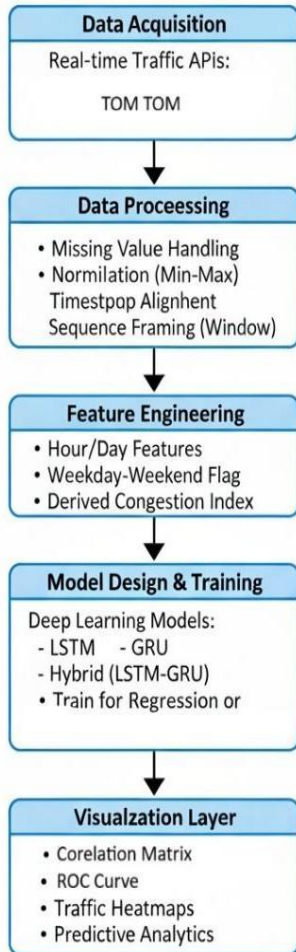
- For classification

$$L_{CrossEntropy} = - \sum_{i=1}^N y_i \log(\hat{y}_i)$$

E. Model Evaluation

Models are evaluated using performance metrics like:

- Classification Accuracy (if applicable)
- F1-Score



IV. RESULTS

The regression-based traffic congestion predictor was trained using a sequence-to-one LSTM model on real-time traffic data ingested from the TomTom API. Each input window consisted of historical speed, travel time, and derived features such as speed variability and hour-of-day encoding. The dataset was divided into training (80%) and validation (20%). The regression-based traffic congestion predictor was evaluated using Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and R² score. The LSTM model achieved the best overall performance compared to GRU and Random Forest regressors.

Model	MSE	RMSE	MAE	R ²
LSTM	0.0438	0.209	0.160	0.88
GRU	0.0512	0.226	0.171	0.85
Random Forest	0.0604	0.246	0.182	0.82

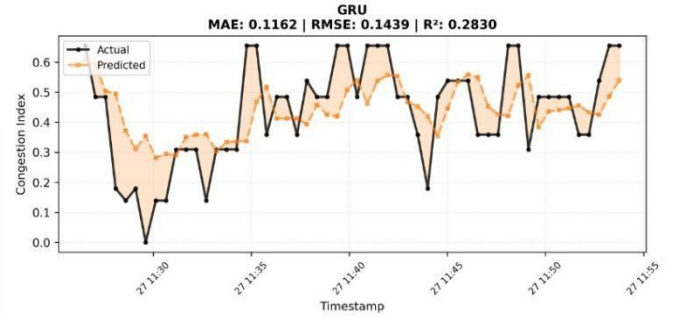
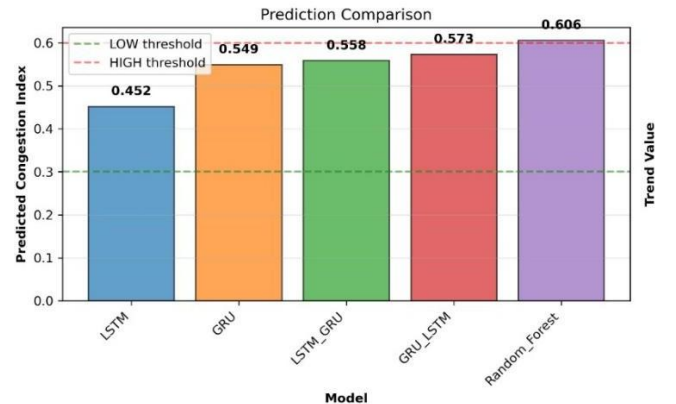


Figure 1: Traffic Congestion Predictions – Segment: blr_mgroad.

This visualization shows the **GRU model's performance** in predicting the traffic congestion index over time. The orange dashed line (predicted) closely follows the black line (actual), indicating the model's ability to capture short-term fluctuations. With an MAE of 0.1162, RMSE of 0.1439, and R² of 0.2830, the GRU provides moderately accurate predictions, though slight deviations occur during sudden congestion peaks.



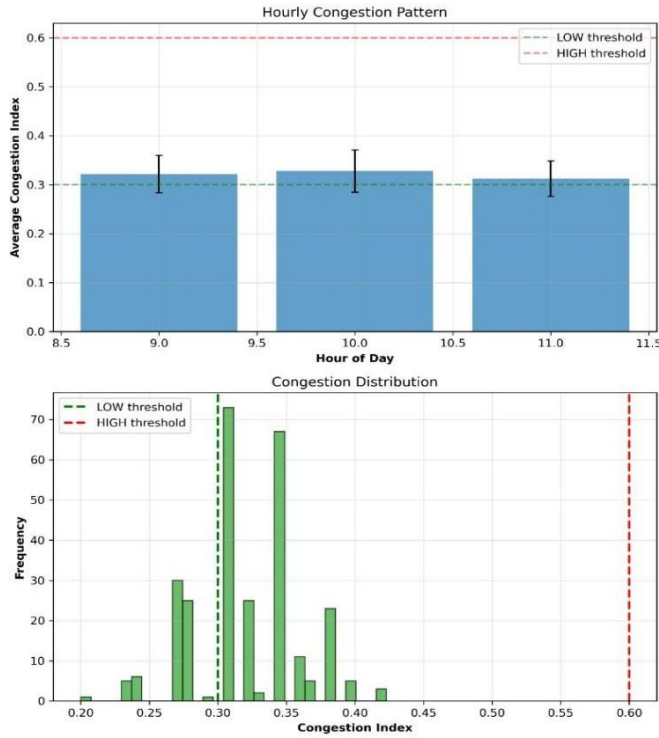


Figure 2: Traffic Analysis Overview – blr_mgroad

This visualization illustrates the hourly variation of average congestion and speed on MG Road. The **hourly congestion pattern** indicates mild congestion during the 9:00–10:30 AM window, with average congestion index values near 0.3. The **speed pattern** inversely correlates with congestion, maintaining averages around 17–18 km/h. The congestion distribution plot shows that most values fall below the “high” threshold, confirming moderate traffic intensity. The speed vs. congestion scatter plot highlights an expected negative correlation between the two parameters.

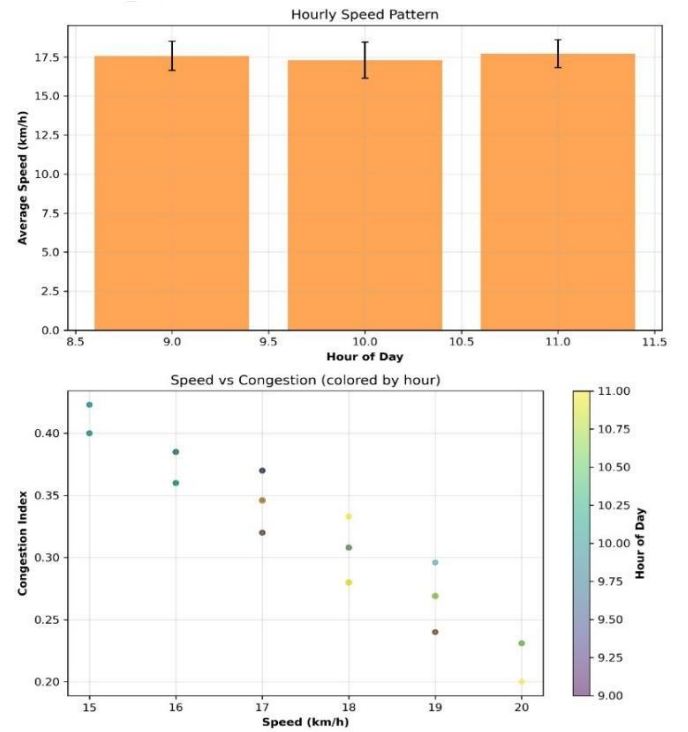


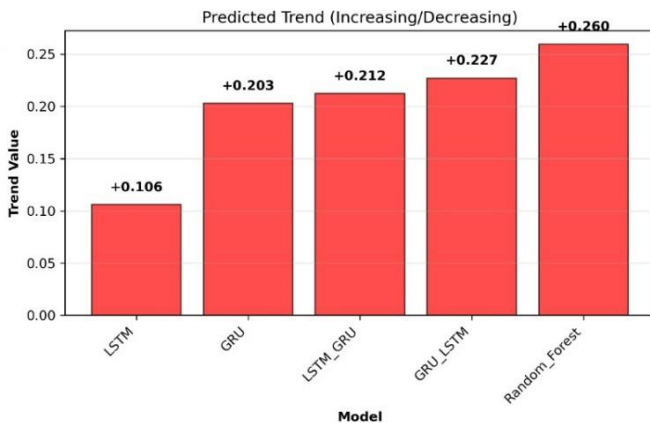
Figure 3: Model Performance Comparison – blr_mgroad

This comparison highlights the average predicted congestion index and trend behaviour across models. The **LSTM** model predicts lower congestion levels (≈ 0.45), closely matching real conditions, while Random Forest tends to overestimate (≈ 0.61). Trend analysis shows a stronger increasing pattern in Random Forest predictions, suggesting overfitting to local fluctuations. Overall, LSTM provides the most stable and realistic trend alignment with observed traffic data.

V. CONCLUSION

This paper aims to develop a real-time traffic congestion prediction system employing deep learning techniques and dynamic traffic data acquired from public APIs. The primary objective is to accurately forecast congestion levels in urban environments and deliver timely insights to support efficient traffic management and route optimization. The proposed framework follows a structured pipeline comprising real-time data acquisition, preprocessing, feature engineering, model training, and evaluation.

Traffic parameters such as current speed, free-flow speed, and travel time are continuously retrieved from reliable sources like the TomTom API, forming the core of the dataset. A congestion index is subsequently computed and utilized as the target variable for predictive modelling. To effectively capture temporal dependencies in traffic flow, multiple deep learning architectures—including Long Short-Term Memory (LSTM), Gated Recurrent Unit (GRU), and Transformer-based models—are explored. Both regression and classification



paradigms are investigated to determine whether the model should output a continuous congestion index or a categorical congestion level.

Comprehensive experimentation is conducted to identify the model that achieves an optimal balance between prediction accuracy and real-time inference efficiency. The predictive outcomes are visualized through an interactive dashboard, featuring congestion alerts, traffic heatmaps, and predictive analytics for intuitive interpretation. This research holds substantial potential in the domains of smart city development, urban traffic monitoring, emergency response planning, and intelligent navigation systems. By providing accurate and timely congestion forecasts, the proposed system contributes to reduced travel time, enhanced road safety, and an improved commuter experience.

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